

INVENTION DISCLOSURE FORM

Applicant Name-Nikhil Saini

Applicant Address-CGC Jhanjeri

1. Particulars of Inventors

Mr./Ms/Dr.	Name (Full)	Department	Designation	Mobile No.	Email	Postal Address
Mr.	Nikhil Saini	CSE-Apex	Student	7814061899	2401002030@cgcjhanjeri.in	CGC Jhanjeri
Mr.	Puneet Goyal	CSE-Apex	Student	9888295938	2401002112@cgcjhanjeri.in	CGC Jhanjeri

2. Provide a brief descriptive title of the invention: “A Database-Driven Rule-Based Web System for Predicting Diseases from Patient-Reported Symptoms.”

3. Summary of the invention: The invention relates to a computer-based web system designed to predict possible diseases using patient-reported symptoms. The proposed system utilizes a rule-based symptom matching approach, powered by a relational MySQL database and a Flask web application, to analyze symptom data and identify diseases associated with user-selected symptoms. Users input their symptoms through a digital web interface by selecting from a predefined list, and the system processes the information using SQL-based symptom overlap scoring to generate probable disease outcomes ranked by match percentage. This approach assists in early detection and preliminary diagnosis, helping individuals seek timely medical consultation. The system aims to improve healthcare accessibility by providing quick, data-driven insights and reducing dependency on manual preliminary assessments while supporting healthcare professionals with an efficient decision-support tool.

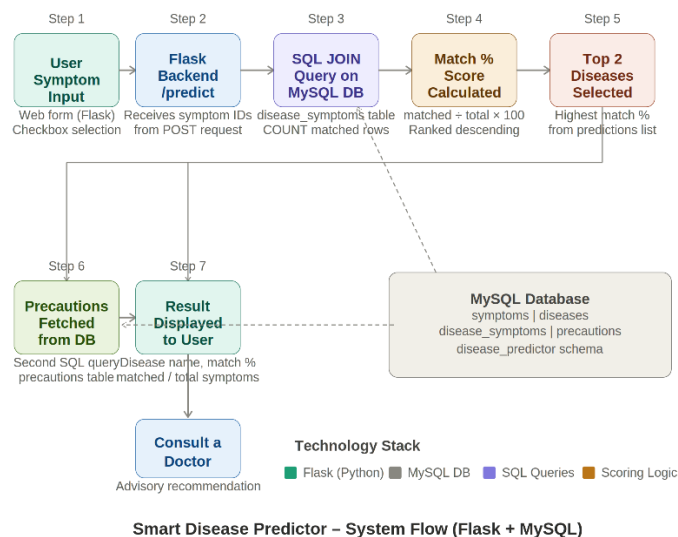
4. Detail description of the invention:

- a. **Problem the invention is solving:** Many individuals experience difficulty in identifying potential diseases based on early symptoms due to limited medical knowledge and delayed access to healthcare professionals. Traditional diagnosis often requires physical consultation, which can be time-consuming, costly, and sometimes inaccessible, especially in remote areas. As a result, early symptoms are frequently ignored or misinterpreted, leading to delayed treatment and worsening health conditions. Additionally, healthcare systems face heavy workloads, making preliminary screening challenging. The invention addresses this problem by providing an automated system that analyzes user-reported symptoms and predicts possible diseases, enabling quick preliminary assessment, improving awareness, and supporting early medical consultation and better healthcare decision-making.
- b. **General Utility/application of the invention:** The invention can be widely used in healthcare applications to assist individuals in identifying possible diseases based on their symptoms. It can be integrated into mobile applications, web platforms, and hospital management systems to provide quick preliminary health assessments. The system helps users gain early awareness of potential health conditions and encourages timely medical consultation. It can also support healthcare professionals by acting as a decision-support tool during initial patient screening. Additionally, the invention is useful in telemedicine services, remote healthcare monitoring, and public health platforms, improving accessibility to basic diagnostic support and enhancing overall efficiency in healthcare delivery systems.
- c. **Advantages of the invention disclosing about the increased efficiency/efficacy :**
 - a. **Faster Preliminary Diagnosis**
The system quickly analyzes user symptoms and predicts possible diseases within seconds, reducing the time required for initial health assessment.
 - b. **Improved Healthcare Accessibility**
It allows users to check possible health conditions anytime through a digital platform, making basic diagnostic support available even in remote areas.

- c. **Support for Medical Professionals**
The system assists doctors by providing preliminary predictions, helping them focus on accurate diagnosis and treatment more efficiently.
 - d. **Early Disease Detection**
By analyzing symptoms at an early stage, the system encourages users to seek timely medical advice, which can help prevent the worsening of diseases.
 - e. **Efficient Symptom-Based Matching Using SQL Database**
The invention uses a relational database with structured symptom-disease mappings and SQL-based overlap scoring to efficiently analyze user-reported symptoms, providing fast and accurate preliminary disease predictions.
- d. **Best way of using the invention as well as possible variants:** The best way to use the invention is through a user-friendly web application built with Flask and MySQL, where users can select their symptoms from a predefined list. The system processes the entered data using SQL-based symptom overlap scoring, querying the database to calculate match percentages between user-selected symptoms and disease-symptom mappings stored in the relational database. The top predicted diseases along with their precautions are displayed to the user. The results provide users with preliminary health insights and recommend seeking professional medical advice if necessary. The invention can also be integrated into hospital management systems, telemedicine platforms, and healthcare applications. Possible variants include incorporating additional patient data such as age, medical history, and lifestyle factors to improve prediction accuracy and provide more personalized health assessments.
- e. **Working of invention along with Drawing, schematics and flow diagrams if required with complete explanations**

System Components :

- 1) **User Symptom Input:**
The user enters their symptoms through a web or mobile interface, selecting from a list or typing the symptoms they are experiencing.
- 2) **Data Preprocessing:**
The system processes the input data by receiving the selected symptom IDs from the web form and converting them into a structured list suitable for SQL query execution against the disease database.
- 3) **Symptom-Disease SQL Matching:**
The selected symptom IDs are matched against the disease_symptoms relational table using a SQL JOIN query. The system counts the number of matching symptoms per disease and calculates a match percentage as $(\text{matched symptoms} / \text{total symptoms of disease}) \times 100$, ranking diseases in descending order of match score.
- 4) **Disease Ranking and Scoring:**
The Flask backend executes the SQL query and retrieves the top two diseases with the highest symptom match percentage. For each predicted disease, a second query fetches associated precautions from the precautions table, providing actionable health guidance.
- 5) **Disease Prediction Generation:**
The system generates a list of probable diseases based on the analyzed symptom data and model predictions.
- 6) **Result Display to User:**
The predicted diseases and related information are displayed to the user through the interface for preliminary health awareness.
- 7) **Recommendation for Medical Consultation:**
The system advises users to consult a healthcare professional for proper diagnosis and treatment based on the predicted results.



5. Have you conducted Primary Patent Search?

NO

6. Existing state-of-the-art and prior arts: (Brief background of the existing knowledge/product/process in the market)

Various digital health systems and applications have been developed to assist in preliminary disease identification based on symptoms. Many existing platforms use rule-based methods or basic database-driven models to analyze symptoms and provide possible disease suggestions. These systems aim to improve healthcare accessibility and support early health awareness.

Some advanced cases have introduced additional features:

- Artificial Intelligence and Deep Learning Integration**
 Advanced systems use AI and deep learning algorithms to analyze complex symptom patterns, improving the accuracy of disease prediction.
- Integration with Electronic Health Records (EHR)**
 Some platforms connect with patient health records to consider medical history, enabling more personalized and accurate predictions.
- Telemedicine Support**
 Modern systems allow users to directly consult doctors through telemedicine services after receiving initial symptom-based predictions.
- Real-Time Health Monitoring**
 Certain applications integrate wearable devices to continuously monitor health parameters such as heart rate or temperature for better disease assessment.

Despite these developments, many systems still face limitations in prediction accuracy, data availability, and accessibility. Continuous improvements in database design and healthcare data integration are required.

The proposed invention builds upon existing technologies by providing an efficient symptom-based prediction system using structured relational databases and SQL scoring, aiming to enhance early disease detection and support faster preliminary healthcare decision-making.

7. List out the known ways about how others have tried to solve the same or similar problems? Indicate the disadvantages of these approaches. In addition, please identify any prior art documentation or other material that explains or provides examples of such prior art efforts.

S. No.	Existing state of art	Drawbacks in existing state of art	Overcome (how your invention is overcoming the drawback)
1	Traditional symptom checker systems based on rule-based algorithms.	These systems often provide limited accuracy because they rely on fixed rules and cannot learn from new data.	The proposed invention also uses a rule-based approach but enhances it with a structured relational MySQL database, SQL-based symptom overlap scoring, and a match percentage calculation that ranks diseases by relevance, providing more structured and reliable results.
2	Online medical information platforms where users manually search symptoms.	Users may misinterpret medical information, leading to incorrect self-diagnosis and confusion.	The invention automatically analyzes symptoms and provides structured disease predictions, reducing the chances of misinterpretation.
3	Healthcare consultation requiring direct visits to hospitals or clinics.	Physical consultations can be time-consuming, costly, and sometimes inaccessible in remote areas.	The invention provides a quick digital symptom-based prediction system that helps users get preliminary health insights before consulting a doctor.

8. List the Technical features and Elements of the invention along with the Description of your invention from start to end.

Component	Function
1. User Interface	Provides a platform where users can enter their symptoms and interact with the system through a web or mobile application.
2. Symptom Input Module	Collects and records the symptoms provided by the user for further processing and analysis.
3. Data Preprocessing Module	Cleans, organizes, and converts the input symptom data into a structured list of symptom IDs suitable for SQL query execution against the relational disease database.
4. Symptom Dataset	Contains structured symptom, disease, disease_symptoms, and precautions data stored in a MySQL relational database, used to match user-reported symptoms against known disease profiles.
5. Machine Learning Model	Executes SQL JOIN queries to count matched symptoms per disease and calculates a match percentage score to rank diseases by relevance to the user's input..
6. Prediction Engine	Generates the ranked list of top probable diseases based on symptom match percentage scores computed from the SQL query results.
7. Result Display Module	Presents the predicted diseases and related recommendations to the user in an understandable format.

Description of the Invention from Start to End

Step : 1. User Symptom Input

- Users access the system via a web or mobile application.
- They select or type the symptoms they are experiencing.

Step : 2. Data Preprocessing

- The system cleans and organizes the entered symptom data.
- Converts the selected symptom checkboxes into a list of integer symptom IDs passed to the Flask backend for SQL query processing.
- Handles missing or ambiguous inputs to ensure accurate analysis.

Step : 3. Symptom-Disease SQL Matching

- Each selected symptom ID is matched against the disease_symptoms table in MySQL using a SQL JOIN query.
- The number of matched symptoms is counted per disease (matched_symptoms) and compared against the total number of symptoms for that disease (total_symptoms).
- A match percentage is calculated as $(\text{matched_symptoms} / \text{total_symptoms}) \times 100$, and diseases are ordered by this score in descending order.

Step : 4. Disease Scoring and Ranking

- The Flask backend retrieves the SQL query results, which include matched symptom counts and total symptom counts for each disease.
- Each disease is scored using the match percentage formula and stored in a predictions list sorted by relevance.
- The top two diseases with the highest match scores are selected for display, along with their associated precautions retrieved from a separate SQL query.

Step : 5. Disease Prediction Generation

- The system generates a ranked list of possible diseases.
- Each predicted disease is accompanied by its match percentage score and the count of matched vs. total symptoms, providing transparency in the prediction.

Step : 6. Result Display and Recommendations

- The results are displayed to the user in a clear, easy-to-understand format.
- The system recommends consulting a healthcare professional for further evaluation.

9. List out the features of your invention which are believed to be new and distinguish them over the closest technology.

Feature	Description
1. Symptom-Based Input	Allows users to enter or select their symptoms through a user-friendly interface for analysis.
2. SQL-Based Symptom Overlap Scoring	Uses SQL JOIN queries against a relational MySQL database to count matched symptoms per disease and rank diseases by match percentage score.
3. Probability Scoring	Assigns a match percentage score to each predicted disease calculated as $(\text{matched symptoms} / \text{total symptoms}) \times 100$, indicating the degree of symptom overlap.
4. User Guidance & Recommendations	Provides advice on consulting healthcare professionals or taking preventive measures based on predictions.
5. Data Preprocessing	Cleans, organizes, and converts user input into a format suitable for analysis, improving prediction reliability.
6. Integration with Health Records	Optional feature to consider patient history, age, or lifestyle factors for more personalized predictions.

Closest Technology	Limitations	How This Invention Is Different
Rule-Based Symptom Checkers	Limited accuracy; cannot learn from new data or evolving symptoms.	Uses a structured relational MySQL database with disease-symptom mappings and SQL-based percentage scoring to provide ranked, data-driven disease predictions with transparency in matched vs. total symptoms.
Online Medical Information Portals	Users interpret data manually, leading to misdiagnosis or confusion.	Automatically analyzes symptoms and provides structured, ranked disease predictions to reduce errors.
Basic Mobile Health Apps	Provide generic advice without considering symptom patterns or probabilities.	Offers probability-based predictions and personalized recommendations for early disease detection.
Telemedicine Platforms	Require prior consultation with doctors; preliminary symptom analysis is limited.	Provides instant symptom-based preliminary disease predictions before professional consultation, saving time and improving accessibility.

10. Has the invention been built or tested or implemented? If so, please provide the particulars of the first time it was successfully built or implemented (when, where, by whom, and evidence of this event including written or on-line pointers to documentary evidence):All testing details\

NO

11. Briefly state when and how you first conceived this idea?

The idea originated while observing how people struggled to recognize potential diseases from common symptoms, often delaying medical consultation. Limited access to healthcare and lack of symptom awareness inspired the concept of a digital system to provide preliminary disease insights efficiently.

The concept was developed using a Flask web application integrated with a MySQL relational database. The system maps user-selected symptoms to disease profiles stored in the database and calculates a match percentage using SQL queries to rank and display the most probable diseases along with their precautions. This approach aims to improve early detection, reduce misdiagnosis, and assist both individuals and healthcare professionals in making informed, timely healthcare decisions, especially in areas with limited medical access.

12. Have you sold, offered for sale, publicly used or published anything related to this invention? If yes, please briefly explain the dates and circumstances. List those individuals to whom you have revealed your invention. Were non-discloser documents signed prior to disclosure in each case? Please state any deadlines of which you may be aware for filing an application on this invention.

No

13. Include any reasons that your invention would not have been obvious to someone of average skill in the art.

NA

14. Additional comments by inventor (if you want to give more details out of scope of this IDF).

NA

15. Drawings/Flowchart/Table

NA